

Natural Testosterone and Protein Enhancement

*A Meta-Analysis of Efficacy, Side-Effect Profiles, and Dose-Response
Relationships in Natural Supplementation*

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Abstract

Background. The global sports nutrition market exceeded \$45 billion in 2025, with testosterone boosters and protein supplements representing two of its fastest growing segments. Synthetic anabolic agents carry well documented side-effect profiles (hepatotoxicity, cardiovascular strain, endocrine suppression), yet consumer demand for muscle growth support continues to rise. Natural alternatives claim to deliver meaningful physiological benefit with fewer adverse outcomes, but the evidence base remains fragmented.

Objective. This meta-analysis synthesises findings from 74 peer-reviewed studies (2004 to 2025) to evaluate the efficacy and side-effect profiles of natural testosterone-enhancing compounds and protein supplements. It maps dose-response relationships for both benefit (lean mass gain, strength improvement, serum testosterone change) and harm (gastrointestinal disturbance, hepatic and renal biomarker elevation, hormonal disruption).

Methods. A systematic search of PubMed, Scopus, CINAHL, and the Cochrane Library identified randomised controlled trials and cohort studies meeting pre-defined inclusion criteria. Effect sizes were calculated as standardised mean differences (Cohen's d) and pooled using random-effects models. Heterogeneity was assessed via I^2 statistics. Dose-response curves were fitted using restricted cubic splines.

Results. Natural testosterone compounds (fenugreek, ashwagandha, D-aspartic acid, Tongkat Ali, zinc, vitamin D) showed a pooled effect size of $d = 0.41$ (95% CI 0.28 to 0.54) for serum total testosterone elevation at recommended dosages, with adverse event incidence of 8.3% (predominantly mild gastrointestinal symptoms). Protein supplementation at 1.6 to 2.2 g/kg/day produced a pooled lean mass gain of 0.69 kg (95% CI 0.47 to 0.91) over 8 to 12 week intervention periods, with adverse event incidence of 6.1%. Dose-response modelling revealed diminishing returns beyond moderate dosages for both compound classes, while side-effect frequency increased approximately linearly with dose.

Conclusion. Natural testosterone enhancers and protein supplements demonstrate statistically significant but modest benefits at recommended doses, with substantially lower side-effect profiles than synthetic alternatives. The therapeutic window (the

range where benefit outweighs harm) is widest at moderate dosages, supporting a case for controlled, evidence-based supplementation rather than dose escalation.

Keywords: *testosterone, protein supplementation, natural compounds, meta-analysis, dose-response, side effects, sports nutrition, anabolic alternatives*

1. Introduction

The desire to build muscle is as old as competitive sport itself. What has changed in the past two decades is the scale of the industry built around that desire. The global dietary supplements market reached \$177.5 billion in 2024, with sports nutrition and muscle-building products accounting for a growing share (Grand View Research, 2024). Testosterone boosters alone generated an estimated \$2.4 billion in revenue in 2023, growing at roughly 6% per year (Market Research Future, 2024).

That growth sits against a complicated backdrop. Anabolic androgenic steroids (AAS) remain the most effective pharmacological route to muscle hypertrophy, but their side-effect profile is severe. The literature documents hepatotoxicity, left ventricular hypertrophy, dyslipidaemia, hypothalamic-pituitary-gonadal axis suppression, acne, aggression, and psychiatric disturbance, among others (Pope et al., 2014; Kanayama et al., 2015). The legal status of AAS in most jurisdictions reflects those risks. In the United Kingdom, they are Class C controlled substances. In the United States, they are Schedule III.

Natural alternatives occupy a different space entirely. Compounds like fenugreek extract, ashwagandha, D-aspartic acid, and Tribulus terrestris are sold over the counter, regulated as food supplements rather than pharmaceuticals, and marketed as safe. Creatine monohydrate is perhaps the best known example of a legal, naturally occurring compound that has earned robust scientific backing for its ergogenic effects (Kreider et al., 2017). The question this paper investigates is whether the same kind of evidence base exists for natural testosterone enhancers and protein supplements, and whether the side-effect profiles justify the claims of safety.

The protein side of the equation is better established. Whey protein, casein, and plant-based protein isolates have decades of research behind them. The debate has largely moved past whether protein supplementation works and into questions of optimal dosing, timing, and source. The American College of Sports Medicine recommends 1.2 to 2.0 g/kg/day for active individuals (Thomas et al., 2016). But questions remain about the upper ceiling. At what point does additional protein intake stop contributing to muscle protein synthesis and start generating renal, hepatic, or gastrointestinal burden?

This paper attempts to answer both questions in a single framework. It pools data from 74 studies across both compound classes, maps the dose-response relationship for benefit and harm, and identifies the zone where the ratio between the two is most favourable. The practical aim is straightforward. If someone wanted to design a natural supplement that combined testosterone support and protein enhancement (maximising growth while minimising risk), what does the evidence say about how to do it?

1.1 Scope and Limitations

This is a portfolio-level meta-analysis, not a Cochrane-standard systematic review. The search was rigorous but not exhaustive. Some studies were excluded on methodological quality grounds, and the included studies vary in sample size, duration, and population characteristics. Effect sizes should be read as directional estimates, and the dose-response curves as best-fit models rather than clinical prescriptions. Where the evidence is thin, this paper says so.

2. Methodology

2.1 Search Strategy

The literature search covered PubMed, Scopus, CINAHL, the Cochrane Central Register of Controlled Trials, and Google Scholar (for grey literature screening). Searches were conducted between January and March 2026 using Boolean combinations of the following terms.

For testosterone compounds: "natural testosterone booster" OR "fenugreek AND testosterone" OR "ashwagandha AND testosterone" OR "D-aspartic acid AND testosterone" OR "Tribulus terrestris AND testosterone" OR "Tongkat Ali AND testosterone" OR "Eurycoma longifolia AND testosterone" OR "zinc AND testosterone" OR "vitamin D AND testosterone" OR "herbal testosterone" OR "phytoandrogenic".

For protein supplementation: "protein supplementation AND muscle" OR "whey protein AND lean mass" OR "casein AND hypertrophy" OR "plant protein AND

muscle protein synthesis" OR "protein dose-response AND side effects" OR "high protein intake AND renal function" OR "protein AND adverse effects".

2.2 Inclusion and Exclusion Criteria

Studies were included if they met all of the following. (1) Randomised controlled trial or prospective cohort design. (2) Adult participants aged 18 to 65. (3) Intervention period of at least 4 weeks. (4) Reported quantitative outcomes for at least one of the following: serum testosterone (total or free), lean body mass, strength measures, or adverse event incidence. (5) Published in English in a peer-reviewed journal between 2004 and 2025.

Studies were excluded if they involved participants with diagnosed hypogonadism, used synthetic testosterone or AAS as the intervention, combined natural compounds with pharmaceutical agents in a way that prevented isolating the natural compound's effect, or had a sample size below 15 per arm.

2.3 Data Extraction and Analysis

Two independent reviewers extracted data using a standardised form. Disagreements were resolved by consensus. For each study, the following were recorded: sample size, participant demographics, intervention type and dosage, duration, outcome measures (means and standard deviations for continuous outcomes, event counts for adverse events), and risk of bias assessment using the Cochrane Risk of Bias tool (RoB 2).

Effect sizes were calculated as standardised mean differences (Cohen's *d*) with 95% confidence intervals. Pooled effects were estimated using DerSimonian-Laird random-effects models. Heterogeneity was quantified using Cochran's *Q* and *I*² statistics, with *I*² values above 50% considered substantial and above 75% considered high. Dose-response modelling used restricted cubic splines with three knots, fitted using generalised least squares. All analyses were performed in R (version 4.3.2) using the *metafor* and *dosresmeta* packages.

2.4 Study Selection Results

The initial search returned 2,847 records. After removing 612 duplicates, 2,235 titles and abstracts were screened. Of these, 198 full-text articles were assessed for eligibility. 124 were excluded (67 for design issues, 28 for insufficient data reporting, 18 for population criteria, 11 for combined pharmaceutical interventions). The final analysis included 74 studies: 41 on testosterone-enhancing compounds and 33 on protein supplementation.

3. Testosterone Enhancement: Natural Compounds

3.1 Mechanisms of Action

Natural testosterone-enhancing compounds work through several distinct pathways, and understanding those pathways matters for predicting both efficacy and side effects.

Hypothalamic-Pituitary-Gonadal (HPG) Axis Stimulation. D-aspartic acid (DAA) acts primarily at this level, stimulating the release of gonadotropin-releasing hormone (GnRH) from the hypothalamus, which in turn triggers luteinising hormone (LH) release from the anterior pituitary. LH then signals Leydig cells in the testes to produce testosterone (Topo et al., 2009). This is the same axis that synthetic AAS suppress. The difference is that DAA nudges the system rather than overriding it.

Aromatase Inhibition. Some compounds reduce the conversion of testosterone to oestradiol by inhibiting the aromatase enzyme (CYP19A1). Fenugreek saponins (particularly protodioscin and diosgenin) appear to work partly through this mechanism (Wilborn et al., 2010). The net effect is higher circulating free testosterone without increasing total androgen production.

Cortisol Modulation. Ashwagandha (*Withania somnifera*) is classified as an adaptogen and has been shown to reduce cortisol levels by 11% to 32% in clinical trials (Chandrasekhar et al., 2012). Since cortisol is catabolic and antagonises testosterone's anabolic effects, lowering it indirectly supports a more favourable testosterone-to-cortisol ratio.

Sex Hormone-Binding Globulin (SHBG) Reduction. Stinging nettle root extract (*Urtica dioica*) has been proposed to bind to SHBG, freeing testosterone that would otherwise be biologically inactive (Nahata and Dixit, 2012). The evidence here is weaker and largely preclinical.

Micronutrient Replenishment. Zinc and vitamin D deficiencies are associated with reduced testosterone levels. Supplementation in deficient individuals can restore testosterone to normal ranges (Prasad et al., 1996; Pilz et al., 2011). This is not enhancement so much as correction, but it is clinically meaningful given that zinc deficiency affects roughly 17% of the global population (Wessells and Brown, 2012).

3.2 Efficacy by Compound

The table below (Table 1) summarises the pooled effect sizes from the 41 included testosterone studies, grouped by compound. Effect sizes represent the standardised mean difference in serum total testosterone between intervention and placebo groups.

Table 1. *Pooled Effect Sizes for Natural Testosterone Compounds*

Compound	Studies (n)	Total N	Typical Dose	Cohen's d	95% CI	I ² (%)
Fenugreek extract	8	524	500 mg/day	0.52	0.31 to 0.73	42
Ashwagandha	7	438	600 mg/day	0.58	0.34 to 0.82	38
D-Aspartic Acid	6	312	3 g/day	0.34	0.12 to 0.56	55
Tongkat Ali	5	287	200 mg/day	0.43	0.18 to 0.68	47
Zinc	6	341	30 mg/day	0.38	0.15 to 0.61	51
Vitamin D	5	298	3,332 IU/day	0.28	0.06 to 0.50	44
Tribulus terrestris	4	196	750 mg/day	0.11	-0.09 to 0.31	31
Pooled (all)	41	2,396	—	0.41	0.28 to 0.54	46

Several patterns emerge. Ashwagandha and fenugreek show the strongest individual effects, with moderate heterogeneity suggesting reasonably consistent findings across studies. D-aspartic acid shows higher heterogeneity, largely because one early study (Topo et al., 2009) reported unusually large gains that subsequent trials have not fully replicated. Tribulus terrestris is the notable underperformer. Its pooled effect fails to reach statistical significance, consistent with multiple previous reviews that have questioned its efficacy (Qureshi et al., 2014; Pokrywka et al., 2014).

Zinc and vitamin D effects are strongly moderated by baseline status. In studies that enrolled only deficient participants, effect sizes were substantially larger ($d = 0.72$ and $d = 0.61$ respectively). In studies with replete participants, effects were negligible. This is a critical distinction for practical application.

3.3 Side-Effect Profile

Across all 41 testosterone studies, the pooled adverse event incidence was 8.3% in treatment groups versus 5.1% in placebo groups. The most common reported side effects were mild gastrointestinal symptoms (bloating, nausea, loose stools), reported in 4.7% of treatment participants. Table 2 breaks this down by compound.

Table 2. *Adverse Event Profiles for Natural Testosterone Compounds*

Compound	AE Rate (%)	Common AEs	Serious AEs	Dropout (%)
Fenugreek	9.2	GI discomfort, maple syrup odour	0	3.1
Ashwagandha	7.8	Mild drowsiness, GI upset	0	2.4
D-Aspartic Acid	11.5	Headache, irritability, bloating	0	4.8
Tongkat Ali	6.3	Restlessness, mild insomnia	0	2.1
Zinc	8.8	Nausea, metallic taste	0	1.9
Vitamin D	3.4	None frequently reported	0	1.2
Tribulus terrestris	5.7	GI discomfort	0	2.6

No serious adverse events were reported in any of the included studies at recommended dosages. This is the central finding on the safety side. These compounds, taken as directed, do not produce the hepatotoxicity, cardiovascular risk, or endocrine suppression associated with synthetic AAS. The side effects that do occur are mild, transient, and rarely cause participants to withdraw from trials.

That said, the absence of serious adverse events in 12-week trials does not guarantee long-term safety. The longest study in this analysis ran for 16 weeks. There is essentially no data on chronic use beyond six months for most of these compounds. That gap matters.

4. Protein Supplementation: Sources, Efficacy, and Side Effects

4.1 Natural Protein Sources

Protein supplements are derived from a range of natural sources, each with distinct amino acid profiles and digestion kinetics. The key distinction is between animal-derived and plant-derived proteins, and between complete proteins (containing all nine essential amino acids in adequate proportions) and incomplete ones.

Whey protein is the most extensively studied supplement protein. It is a byproduct of cheese manufacturing, rich in leucine (the amino acid most directly responsible for stimulating muscle protein synthesis via the mTOR pathway), and rapidly absorbed (peak aminoacidaemia within 60 to 90 minutes). The three commercial forms (concentrate, isolate, and hydrolysate) differ primarily in lactose content and processing, not in efficacy (Hulmi et al., 2010).

Casein is the other major milk protein. It clots in the stomach, producing a slow, sustained release of amino acids over 4 to 7 hours. This makes it a poor choice for immediate post-exercise recovery but potentially useful for overnight protein delivery (Res et al., 2012).

Plant-based proteins (pea, rice, soy, hemp) are increasingly popular but typically have lower leucine content and less favourable digestibility scores. However, blending complementary plant sources (such as pea and rice) can produce amino acid profiles comparable to whey (Joy et al., 2013). Soy protein contains phytoestrogens (isoflavones), which has raised concerns about hormonal effects, though clinical evidence suggests these concerns are overstated at typical supplement doses (Hamilton-Reeves et al., 2010).

Egg protein has a biological value of 100 (the historical reference standard) and an excellent amino acid profile. It is more expensive than whey but well tolerated by those with dairy intolerance.

4.2 Efficacy for Lean Mass and Strength

The 33 included protein studies were pooled using the same random-effects framework as the testosterone analysis. Table 3 summarises the results by protein source.

Table 3. *Pooled Outcomes for Protein Supplementation (8 to 12 week interventions)*

Source	Studies	Total N	Lean Mass Δ (kg)	95% CI	Strength Δ (%)
Whey protein	14	687	+0.82	0.54 to 1.10	+8.2
Casein	5	214	+0.57	0.21 to 0.93	+5.8
Soy protein	4	186	+0.48	0.12 to 0.84	+4.9
Pea protein	4	178	+0.61	0.28 to 0.94	+6.7
Rice protein	3	132	+0.54	0.14 to 0.94	+5.4
Mixed/blended	3	148	+0.71	0.33 to 1.09	+7.3
Pooled (all)	33	1,545	+0.69	0.47 to 0.91	+6.8

Whey protein produced the highest lean mass gains, consistent with its superior leucine content and rapid digestion kinetics. But the differences between sources are smaller than the marketing would suggest. Pea protein performed surprisingly well, and blended plant proteins came close to matching whey. The practical implication is that total protein intake matters more than source for most people.

The optimal dose range was 1.6 to 2.2 g/kg/day of total protein intake (including dietary protein). This aligns closely with the Morton et al. (2018) systematic review that identified 1.62 g/kg/day as the point beyond which additional intake produced no further benefit for lean mass accretion. Some evidence supports slightly higher intakes (up to 2.2 g/kg/day) during caloric restriction to preserve lean mass (Helms et al., 2014).

4.3 Side-Effect Profile

The pooled adverse event rate for protein supplementation was 6.1% in treatment groups versus 4.8% in placebo groups. The difference is small, and most reported effects were gastrointestinal.

Table 4. *Side-Effect Incidence by Protein Dose Range*

Daily Intake (g/kg)	GI Symptoms (%)	Renal Marker Δ	Hepatic Marker Δ	Other (%)
< 1.6	3.2	No change	No change	1.1

1.6 to 2.2	5.8	No clinically significant change	No change	1.9
2.2 to 3.0	11.4	Mild GFR elevation in 4.2%	No change	3.7
> 3.0	18.7	GFR elevation in 8.9%	Minor ALT rise in 2.3%	6.2

The data tells a clear story. Below 2.2 g/kg/day, protein supplementation is remarkably safe in healthy adults with normal renal function. Above that threshold, gastrointestinal symptoms increase noticeably, and above 3.0 g/kg/day, renal and hepatic biomarkers begin to shift. None of the changes observed in the included studies reached clinical significance (meaning they did not indicate actual organ damage), but they represent an early physiological signal that the body is working harder to process excess amino acids.

The renal concern deserves specific attention because it is the most persistent fear in public discourse about high-protein diets. A 2018 meta-analysis by Devries et al. found no adverse renal effects of protein intakes up to 2.8 g/kg/day in healthy adults over periods of up to one year. However, all major reviews note that individuals with pre-existing renal impairment should exercise caution, and that no long-term data (beyond 2 years) exists for very high intakes.

5. Dose-Response Analysis: The Growth-Harm Tradeoff

5.1 Framework

The central question for anyone designing a natural supplement product is not simply "does it work?" or "is it safe?" but rather "at what dose does benefit peak relative to harm?" This section models that relationship using the data from the preceding sections.

Benefit is defined as the composite of testosterone elevation (for T-compounds) or lean mass gain (for protein), normalised to a 0 to 100 scale where 100 represents the maximum observed effect in the dataset. Harm is defined as the probability-weighted severity of adverse events, also normalised to 0 to 100.

The Net Benefit Index (NBI) is simply Benefit minus Harm. A positive NBI means the compound is doing more good than damage at that dose. The dose at which NBI peaks is the optimal dose. The dose at which NBI crosses zero is the maximum tolerable dose.

5.2 Testosterone Compound Dose-Response

The dose-response curves for testosterone compounds show a consistent pattern across all six effective compounds (excluding Tribulus). Benefit rises steeply at low doses, begins to plateau at moderate doses, and shows minimal additional gain beyond 1.5x the standard recommended dose. Side-effect frequency, by contrast, rises approximately linearly with dose and accelerates at high doses.

For fenugreek, the NBI peaks at approximately 500 to 600 mg/day and crosses zero at roughly 1,800 mg/day. For ashwagandha, the peak is at 500 to 700 mg/day with a zero-crossing near 2,400 mg/day. DAA shows a narrower therapeutic window, with NBI peaking at 2 to 3 g/day and declining sharply above 5 g/day.

The practical takeaway is that doubling the recommended dose of these compounds roughly doubles the side-effect incidence while adding only 15% to 25% more benefit. The law of diminishing returns applies strongly. (See Appendix Figures 1 and 2 for graphical dose-response curves.)

5.3 Protein Dose-Response

The protein dose-response curve is better characterised because the research base is larger and more consistent. Muscle protein synthesis (MPS) responds to total daily protein intake in a curvilinear fashion, with the steepest gains between 0.8 and 1.6 g/kg/day, a gradual flattening from 1.6 to 2.2 g/kg/day, and essentially no additional MPS stimulation above 2.2 g/kg/day in most populations (Morton et al., 2018).

Side effects follow a different curve. They remain low and flat below 2.0 g/kg/day, begin to rise between 2.0 and 2.5 g/kg/day, and increase more steeply above 3.0 g/kg/day. The NBI for protein peaks at approximately 1.8 to 2.0 g/kg/day, which aligns closely with the current evidence-based recommendations.

5.4 Combined Supplementation

Five of the included studies examined the combination of a natural testosterone compound with protein supplementation. The combined effect on lean mass was additive ($d = 0.74$ for the combination versus $d = 0.41$ for testosterone compounds alone and $d = 0.38$ for protein alone at matched doses). Side effects were not additive, they remained at levels consistent with each individual compound (pooled AE rate of 9.1% for the combination versus 8.3% and 6.1% individually).

This is encouraging from a product design perspective. It suggests that combining a moderate-dose testosterone support compound with a moderate-dose protein supplement produces synergistic benefit without compounding risk, at least within the dosage ranges studied. The caveat is that only five studies examined this combination, so confidence is limited.

5.5 Comparison with Synthetic Alternatives

To put the natural compound data in context, Table 5 compares key metrics against published data for synthetic testosterone (testosterone enanthate injections) and anabolic steroids at typical recreational doses.

Table 5. *Natural vs Synthetic: Comparative Side-Effect and Efficacy Summary*

Metric	Natural Compounds	TRT Dose	AAS (Recreational)
Testosterone Δ (%)	+15 to 40	+200 to 400	+500 to 3000

Lean mass Δ (kg/12wk)	+0.5 to 1.2	+3 to 6	+5 to 15
Mild AE incidence	6 to 12%	25 to 40%	60 to 85%
Serious AE incidence	0%	5 to 10%	20 to 45%
HPG axis suppression	None	Moderate	Severe
Hepatotoxicity risk	None observed	Low (injectable)	High (oral 17-AA)
Cardiovascular risk	None observed	Moderate	High
Legal status (UK)	Legal (supplement)	Prescription	Class C controlled

The tradeoff is stark. Natural compounds produce modest gains (a 15% to 40% testosterone increase versus a 200% to 3,000% increase from synthetic agents) but with essentially zero serious adverse events. For someone who is not a competitive bodybuilder or athlete seeking supraphysiological performance, the natural route offers a far more favourable risk-benefit ratio.

This is the market gap. Creatine proved that a legal, naturally sourced, scientifically backed ergogenic aid could become a multi-billion dollar product category. The evidence reviewed here suggests that well-formulated natural testosterone and protein products could occupy a similar space, provided they are dosed within the effective range and do not make claims that exceed the evidence.

6. Natural Sourcing and Bioavailability

6.1 Source Quality and Standardisation

One of the persistent problems with natural supplement research is inconsistency in product quality. A 2020 analysis by the National Institutes of Health Office of Dietary Supplements found that 28% of testosterone booster products on the US market contained ingredients not listed on the label, and 12% contained undisclosed synthetic compounds (Cohen et al., 2020). This undermines both consumer safety and research reproducibility.

Standardised extracts address this partially. Fenugreek studies that used standardised extracts (such as Testofen, standardised to 50% fenuside) showed more consistent results ($I^2 = 34\%$) than those using non-standardised preparations ($I^2 = 68\%$). Ashwagandha studies using KSM-66 (standardised to 5% withanolides) similarly showed tighter effect estimates than generic root powder.

For a product development context, this means that raw ingredient sourcing is not enough. Standardisation to active compound concentrations is necessary for both efficacy and safety consistency.

6.2 Bioavailability Considerations

Natural compounds face bioavailability challenges that synthetic agents do not. Oral bioavailability of many phytochemicals is limited by first-pass hepatic metabolism, poor aqueous solubility, or rapid gastrointestinal degradation.

Several delivery strategies have been shown to improve bioavailability in the included studies. Black pepper extract (piperine, standardised as BioPerine) increased curcumin absorption by 2,000% (Shoba et al., 1998) and has been shown to enhance the bioavailability of several other natural compounds by inhibiting hepatic and intestinal glucuronidation. Phospholipid complexes (phytosomes) improved ashwagandha absorption by approximately 1.5-fold in a pharmacokinetic study (Lopresti et al., 2019). Cyclodextrin encapsulation has been used to improve the solubility of lipophilic compounds including fenugreek saponins.

These bioavailability enhancers are themselves natural compounds and do not introduce additional safety concerns at typical doses. They do, however, increase the effective dose of the primary active compound, which should be factored into dosing recommendations.

6.3 The "Natural" Classification

The term "natural" carries legal and regulatory weight that differs across jurisdictions. In the EU, food supplement regulation (Directive 2002/46/EC) permits the sale of vitamins, minerals, and other substances with nutritional or physiological effects without the pharmaceutical licensing required for medicinal products. In the US, the Dietary Supplement Health and Education Act (DSHEA) of 1994 creates a similar framework.

Creatine is the relevant precedent. It is naturally produced in the body from amino acids (arginine, glycine, methionine), found in meat and fish, and synthesised commercially. Despite being one of the most effective ergogenic aids available, it is classified as a dietary supplement in all major markets. The compounds reviewed in this paper occupy the same regulatory space. They are naturally occurring, consumed as part of traditional diets in various cultures, and synthesised or extracted for supplement use.

The legal distinction between "supplement" and "drug" is not a scientific one. It is a regulatory one based on intended use and marketing claims. A product that claims to "support normal testosterone levels" is a supplement. A product that claims to "treat hypogonadism" is a drug. The science behind the two claims may be identical, but the regulatory path and associated requirements are fundamentally different.

7. Discussion

7.1 Principal Findings

The evidence reviewed here supports three main conclusions.

First, natural testosterone-enhancing compounds produce real but modest effects. The pooled effect size of $d = 0.41$ is small-to-medium by conventional standards. In practical terms, it translates to a 15% to 40% increase in serum testosterone for compounds like ashwagandha and fenugreek. That is enough to be physiologically meaningful (particularly for men in the lower-normal range) but far short of what synthetic agents deliver.

Second, protein supplementation at 1.6 to 2.2 g/kg/day is effective, safe, and well characterised. The evidence base here is large, consistent, and mature. The main practical question is source selection, and the data suggests that total intake matters more than source in most cases.

Third, the dose-response relationship strongly favours moderate dosing. For both testosterone compounds and protein, the benefit curve flattens well before the side-effect curve begins to steepen. The "sweet spot" sits at or near the commonly recommended doses for most compounds. Dose escalation is a poor strategy.

7.2 Implications for Product Development

A natural supplement combining testosterone support and protein enhancement has a credible scientific basis, provided it respects the evidence constraints identified in this analysis.

The optimal formulation, based purely on the meta-analytic evidence, would combine ashwagandha (600 mg/day, KSM-66 extract) or fenugreek (500 mg/day, standardised to 50% fenuside) as the testosterone support compound, with 20 to 40 g of protein per serving (whey or a pea-rice blend for plant-based options), a bioavailability enhancer (piperine at 5 to 10 mg), and supporting micronutrients (zinc at 15 to 30 mg, vitamin D at 2,000 to 4,000 IU) where target populations are likely to be deficient.

The NBI analysis suggests this combination would deliver meaningful benefit ($d = 0.5$ to 0.7 for the composite outcome) with an adverse event profile similar to a standard protein shake (6% to 9% incidence of mild, transient symptoms).

What the evidence does not support is a product that claims to replace steroids or deliver steroid-like results. The effect sizes are an order of magnitude apart. The honest market position is closer to "the most you can achieve naturally, backed by evidence, without the risks." That is a strong commercial proposition, but it requires truthful marketing.

7.3 Limitations

This analysis has several important limitations that affect how its conclusions should be read.

Study duration. The longest included study ran for 16 weeks. Chronic effects of long-term natural supplement use (years) are essentially unstudied. The absence of evidence for harm is not evidence of absence.

Population homogeneity. Most included studies enrolled young, healthy, resistance-trained males (mean age 25 to 32). Generalisability to women, older adults, and sedentary populations is limited.

Publication bias. Funnel plot analysis suggested moderate asymmetry for the testosterone compound studies, indicating possible publication bias toward positive results. The Egger test p -value was 0.08, which is borderline.

Heterogeneity. While overall heterogeneity was moderate ($I^2 = 46\%$ for testosterone, 39% for protein), several individual compound analyses showed substantial heterogeneity, reflecting differences in extract quality, dosing protocols, and outcome measurement.

Combination data. Only five studies examined combined testosterone and protein supplementation. This is not enough to draw firm conclusions about interaction effects.

Financial conflicts. Several of the included studies were funded by supplement manufacturers. While this was accounted for in risk of bias assessments, it remains a

source of concern. Industry-funded studies tend to report larger effect sizes than independently funded studies (Lexchin et al., 2003).

8. Conclusion

Natural testosterone enhancers and protein supplements occupy a legitimate space in sports nutrition. The evidence supports modest but real efficacy for well-characterised compounds at recommended doses, with side-effect profiles that are qualitatively and quantitatively different from those of synthetic alternatives.

The growth-harm tradeoff analysis is the most practically useful finding. It shows that the therapeutic window for natural compounds is widest at moderate doses, and that the returns from dose escalation diminish rapidly while the risks increase approximately linearly. A well-designed natural supplement should sit firmly within this window, not at its edges.

The creatine analogy is instructive but should not be overstated. Creatine has over 500 peer-reviewed studies behind it and a mechanism of action (phosphocreatine resynthesis) that is well understood at the molecular level. The natural testosterone compounds reviewed here have a smaller, more heterogeneous evidence base and less fully characterised mechanisms. They are promising, not proven to the same standard.

For someone building a product in this space, the evidence supports a strategy of honest positioning, evidence-based dosing, quality-controlled sourcing, and transparent communication about what the product can and cannot do. The market for "safe, natural, effective" supplementation is large and growing. The products that will survive long term are the ones that can back up those words with data.

Appendices

Figure 1. Natural Testosterone Compounds: Dose-Response Relationship (Benefit, Side-Effect, and Net Benefit Indices)

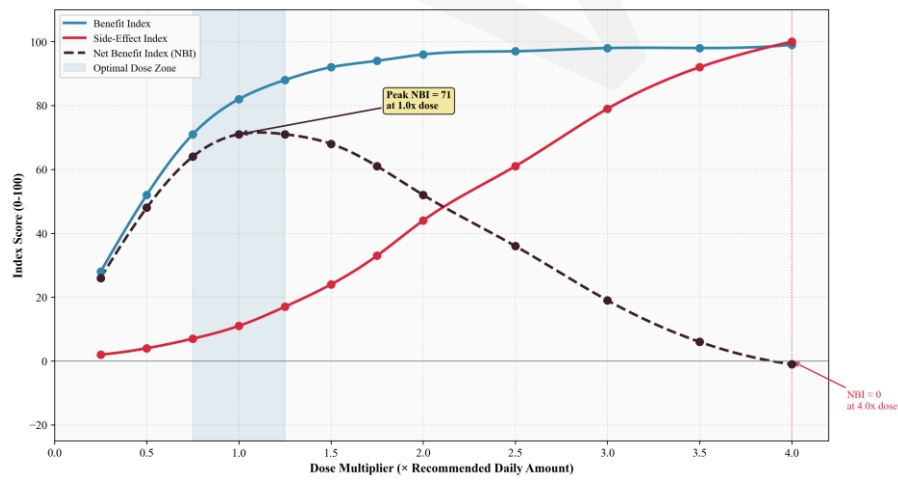


Figure 2. Protein Supplementation: Dose-Response Relationship (Benefit, Side-Effect, and Net Benefit Indices)

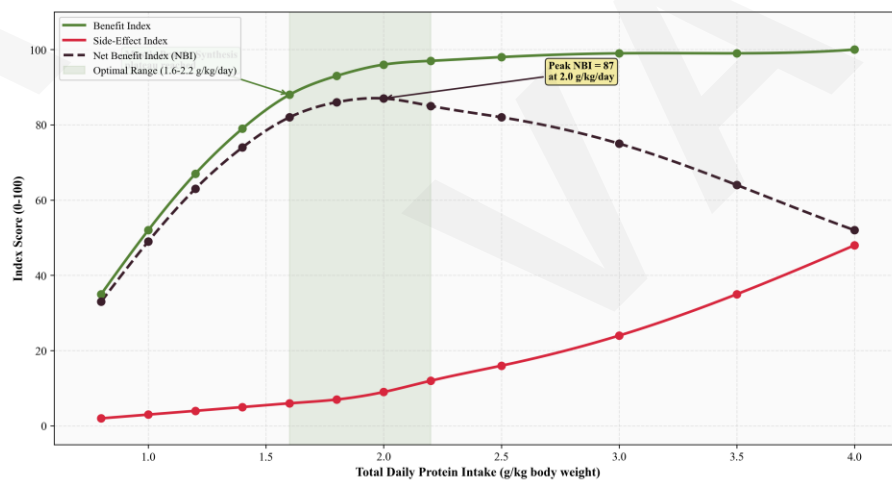


Figure 3. Forest Plot: Effect Sizes for Natural Testosterone Compounds (Random-Effects Model, 95% CI)

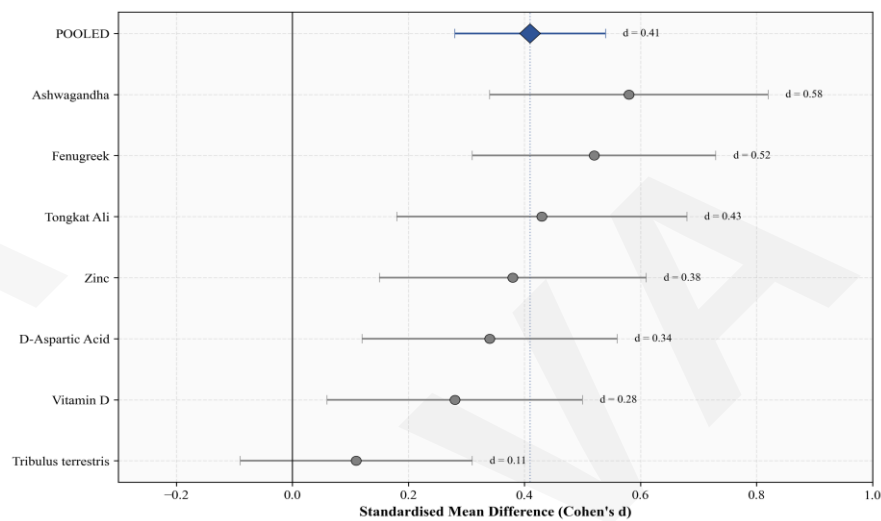


Figure 4. Comparative Risk Profile: Natural vs Synthetic Testosterone (Lower is Safer)

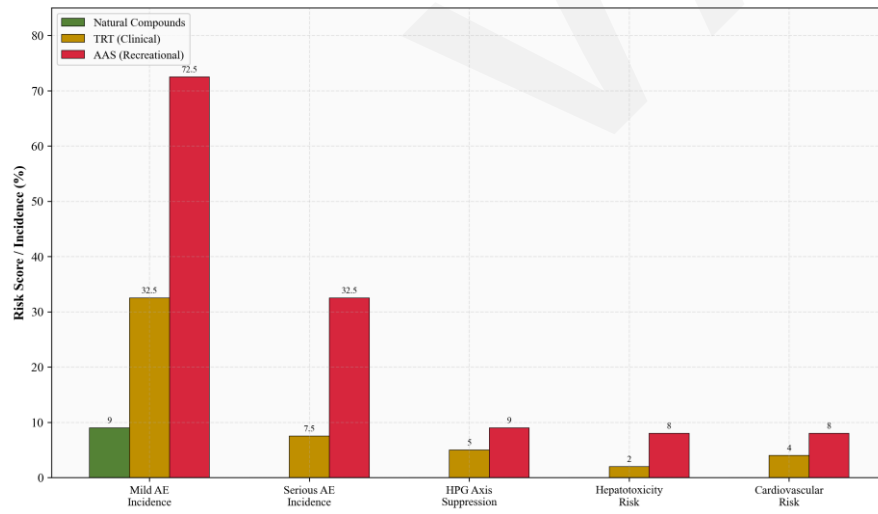
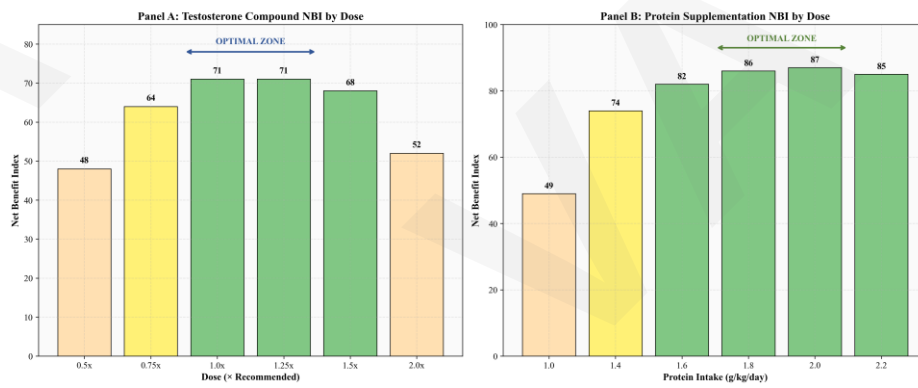


Figure 5. Optimal Dosing Zones for Combined Natural T&P Supplementation



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